

Exercise (1):

1) $\int a dx = ax + c$, $F(x) = ax + c$ and $f(x) = a$

$$\lim_{x \rightarrow x_0} \frac{F(x) - F(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{ax + c - ax_0 - c}{x - x_0} = \lim_{x \rightarrow x_0} \frac{a(x - x_0)}{x - x_0} = a = f(x_0) \text{ so: } \int a dx = ax + c$$

2) $\int x dx = \frac{x^2}{2} + c$, $F(x) = \frac{x^2}{2} + c$ and $f(x) = x$

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{F(x) - F(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{\frac{x^2}{2} + c - \frac{x_0^2}{2} - c}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\frac{x^2 - x_0^2}{2}}{x - x_0} = \lim_{x \rightarrow x_0} \frac{1}{2} \cdot \frac{x^2 - x_0^2}{x - x_0} = \lim_{x \rightarrow x_0} \frac{1}{2} \cdot \frac{(x - x_0)(x + x_0)}{x - x_0} \\ &= \frac{1}{2} \cdot 2x_0 = x_0 = f(x_0) \text{ so: } \int x dx = \frac{x^2}{2} + c \end{aligned}$$

3) $\int x^m dx = \frac{x^{m+1}}{m+1} + c$ ($m \in \mathbb{N}$), $F(x) = \frac{x^{m+1}}{m+1} + c$ and $f(x) = x^m$

$$\lim_{x \rightarrow x_0} \frac{F(x) - F(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\frac{x^{m+1}}{m+1} - \frac{x_0^{m+1}}{m+1}}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\frac{x^{m+1} - x_0^{m+1}}{m+1}}{x - x_0} = \lim_{x \rightarrow x_0} \frac{1}{m+1} \cdot \frac{x^{m+1} - x_0^{m+1}}{x - x_0}$$

$$= \lim_{x \rightarrow x_0} \frac{1}{m+1} \cdot \sum_{k=0}^{m-1} x^{m-k-1} \cdot x_0^k = \frac{1}{m+1} \cdot \sum_{k=0}^m x_0^m = \frac{1}{m+1} x_0^m (m+1) = x_0^m = f(x_0)$$

so: $\int x^m dx = \frac{x^{m+1}}{m+1} + c$

4) $\int \frac{dx}{x^2} = \frac{-1}{x} + c$ ($x \neq 0$), $F(x) = \frac{-1}{x} + c$ and $f(x) = \frac{1}{x^2}$

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{F(x) - F(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{\frac{-1}{x} + c - \frac{-1}{x_0} - c}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\frac{-x_0 + x}{x \cdot x_0}}{x - x_0} = \lim_{x \rightarrow x_0} \frac{-x_0 + x}{x \cdot x_0} \cdot \frac{1}{-x_0 + x} = \lim_{x \rightarrow x_0} \frac{1}{x \cdot x_0} \\ &= \frac{1}{x_0^2} = f(x_0) \text{ so: } \int \frac{dx}{x^2} = \frac{-1}{x} + c \end{aligned}$$

5) $\int \frac{dx}{2\sqrt{x}} = \sqrt{x} + c$ ($x \neq 0$), $F(x) = \sqrt{x} + c$ and $f(x) = \frac{1}{2\sqrt{x}}$

$$\begin{aligned} \lim_{x \rightarrow x_0} \frac{F(x) - F(x_0)}{x - x_0} &= \lim_{x \rightarrow x_0} \frac{\sqrt{x} + c - \sqrt{x_0} - c}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\sqrt{x} - \sqrt{x_0}}{x - x_0} \cdot \frac{\sqrt{x} + \sqrt{x_0}}{\sqrt{x} + \sqrt{x_0}} = \lim_{x \rightarrow x_0} \frac{x - x_0}{(x - x_0)(\sqrt{x} + \sqrt{x_0})} = \lim_{x \rightarrow x_0} \frac{1}{\sqrt{x} + \sqrt{x_0}} = \frac{1}{2\sqrt{x_0}} \\ &= f(x_0) \text{ so: } \int \frac{dx}{2\sqrt{x}} = \sqrt{x} + c \end{aligned}$$

7) $\int \frac{dx}{x} = \ln|x| + c$ ($x \neq 0$), $F(x) = \ln|x| + c$ and $f(x) = \frac{1}{x}$

$$\lim_{x \rightarrow x_0} \frac{F(x) - F(x_0)}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\ln|x| + c - \ln|x_0| - c}{x - x_0} = \lim_{x \rightarrow x_0} \frac{\ln|x| - \ln|x_0|}{x - x_0}$$

↳ case (1): when $x > 0$, we put $x = e^t$: $x \rightarrow x_0 \Leftrightarrow t \rightarrow t_0$

$$\lim_{x \rightarrow x_0} \frac{\ln|x| - \ln|x_0|}{x - x_0} = \lim_{t \rightarrow t_0} \frac{t - t_0}{e^t - e^{t_0}} = \frac{1}{e^{t_0}} = \frac{1}{x_0}$$

↳ case (2): when $x < 0$, we put $-x = e^t$

$$\lim_{x \rightarrow x_0} \frac{\ln|x| - \ln|x_0|}{x - x_0} = \lim_{t \rightarrow t_0} \frac{t - t_0}{e^t - e^{t_0}} = \frac{1}{e^{t_0}} = \frac{1}{x_0} \quad . \text{So: } \int \frac{dx}{x} = \ln|x| + C$$

$$8) \int e^x dx = e^x + C \quad , F(x) = e^x + C \quad \text{and } f(x) = e^x$$

$$\lim_{x \rightarrow x_0} \frac{\cancel{e^x + C} - \cancel{e^{x_0} + C}}{x - x_0} = \lim_{x \rightarrow x_0} \frac{e^x - e^{x_0}}{x - x_0} \quad . \text{Let } n = x - x_0, \text{ we get: } \lim_{n \rightarrow 0} \frac{e^x - e^{x_0}}{n} \cdot e^n$$

Exercise (2):

$$1) \int x^2 \ln(x) dx = \frac{1}{3} x^3 \cdot \ln(x) - \frac{1}{3} \int x^3 \cdot \frac{1}{x} dx = \frac{1}{3} x^3 \ln(x) - \frac{1}{3} \int x^2 dx$$

$$u = \ln(x) \quad | \quad u' = 1/x \quad dx \quad \Rightarrow \quad = \frac{1}{3} x^3 \ln(x) - \frac{1}{3} \left(\frac{1}{3} x^3 \right) = x^3 \left(\frac{\ln(x)}{3} - \frac{1}{9} \right) + C$$

$$v = \frac{1}{3} x^3 \quad | \quad v' = x^2 dx$$

$$2) \int \ln(x) dx = x \ln(x) - \int \frac{1}{x} \cdot x dx = x \ln(x) - \int 1 dx + C = x \ln(x) - x + C$$

$$u = x \quad | \quad u' = 1 \quad dx$$

$$v = \ln(x) \quad | \quad v' = 1/x \quad dx$$

$$3) \int (\ln(x))^2 dx = x(\ln(x))^2 - 2 \int \ln(x) \cdot \frac{1}{x} \cdot x dx = x(\ln(x))^2 - 2x \ln(x) + 2x + C$$

$$u = (\ln(x))^2 \quad | \quad u' = 2 \cdot \ln(x) \cdot \frac{1}{x} dx$$

$$v = x \quad | \quad v' = 1 dx$$

$$4) \int \cos(x) \cdot \exp(x) dx = \exp(x) \cdot \cos(x) + \int \sin(x) \cdot \exp(x)$$

$$u = \exp(x) \quad | \quad u' = \exp(x) \quad \Rightarrow \quad \begin{array}{l} p = \exp(x) \quad | \quad p' = \exp(x) \\ q = \sin(x) \quad | \quad q' = \cos(x) \end{array} \quad \text{so: } \int \sin(x) \cdot \exp(x) = \exp(x) \cdot \sin(x) - \int \cos(x) \cdot \exp(x)$$

$$\int \cos(x) \cdot \exp(x) dx = \exp(x) \cdot \cos(x) + [\exp(x) \cdot \sin(x) - \int \cos(x) \cdot \exp(x)] =$$

$$\Rightarrow 2 \int \exp(x) \cdot \cos(x) dx = \exp(x) \cdot \cos(x) + \sin(x) \cdot \exp(x) + C$$

$$\Rightarrow \int \cos(x) \cdot \exp(x) dx = \frac{1}{2} e^x (\cos(x) + \sin(x)) + C$$

Exercise (3):

$$1) A = \int \frac{\sinh(x)}{1 + 4 \cosh^2(x)} dx \quad \text{we take: } u = \cosh(x), \text{ so: } du = \sinh(x) dx \text{ then:}$$

$$A = \int \frac{1}{1 + (2u)^2} du = \frac{1}{2} \int \frac{1}{1 + t^2} dt = \frac{1}{2} \arctan(2u) + C, \text{ by taking: } t = 2u, \text{ so:}$$

$$dt = 2 du \text{ we replace: } A = \frac{1}{2} \arctan(2 \cosh(x)) + C$$

$$2) B = \int \sqrt{\cos(x)} \cdot \sin(x) dx \quad \text{Let: } u = \cos(x) \text{ and: } du = -\sin(x) dx \text{ then:}$$

$$B = - \int u^{1/2} du = - \frac{u^{3/2}}{3/2} + C = - \frac{2 \cdot u^{3/2}}{3} + C \quad \text{so: } B = - \frac{2}{3} (\cos(x))^{3/2} + C$$

$$3) C = \int \frac{1}{\sqrt{1-x^2} \cdot \arcsin(x)} dx \quad \text{Let: } t = \arcsin(x), \text{ so: } dt = \frac{1}{\sqrt{1-x^2}} dx \text{ then:}$$

$$\int \frac{1}{t} dt = \ln|t| + C \quad \text{so: } C = \ln|\arcsin(x)| + C$$

$$4) D = \int \frac{1}{x \ln(x) \ln(\ln(x))} dx. \text{ let: } u = \ln(\ln(x)), \text{ so: } du = \frac{1}{x \ln(x)} \cdot \text{then:}$$

$$D = \int \frac{du}{u} = \ln|u| + C. \text{ we replace: } D = \ln|\ln(\ln(x))| + C.$$

Exercise (4):

$$1) A = \int \frac{2x+1}{x^2+x+1} dx. \text{ let: } u = x^2+x+1 \Rightarrow du = 2x+1 dx. \text{ so: } A = \int \frac{du}{u} = \ln|u| + C$$

$$\Rightarrow A = \ln(|x^2+x+1|) + C$$

$$2) B = \int_0^2 \frac{x}{x^4+16} dx. \text{ let: } u = x^2 \Rightarrow du = 2x \cdot dx. \text{ so: } B = \frac{1}{2} \int \frac{1}{u^2+16} du = \frac{1}{2} \int \frac{1}{u^2+u^2} du$$

$$\text{we have: } \int \frac{1}{u^2+a^2} du = \frac{1}{a} \arctan\left(\frac{u}{a}\right) + C. \text{ then:}$$

$$B = \frac{1}{2} \times \frac{1}{4} \arctan\left(\frac{u}{4}\right) + C = \frac{1}{8} \arctan\left(\frac{x^2}{4}\right) + C.$$

$$3) C = \int (x+1) \cdot \sinh(x) dx. \text{ let: } u = \cosh(x) \Rightarrow du = \sinh(x) \cdot dx$$

$$v = (x+1) \Rightarrow dv = x \cdot dx$$

$$\text{then: } C = (x+1) \cdot \cosh(x) - \int x \cdot \cosh(x) \cdot dx \Rightarrow C = (x+1) \cdot \cosh(x) - \sinh(x) + C.$$

$$4) D = \int \frac{x^3}{\sqrt{x^2+1}} \cdot dx$$

$$\text{by I.B.P: let: } u = \sqrt{x^2+1} \Rightarrow du = \frac{x}{\sqrt{x^2+1}} dx. \text{ so: } D = x^2 \sqrt{x^2+1} - \int 2x \sqrt{x^2+1} \cdot dx$$

$$v = x^2 \Rightarrow dv = 2x \cdot dx = x^2 \sqrt{x^2+1} - \int \sqrt{u} \cdot du, \text{ by taking: } u = x^2+1 \Rightarrow du = 2x \cdot dx$$

$$D = x^2 \sqrt{x^2+1} - \frac{2}{3} (x^2+1)^{3/2} + C.$$

$$\text{by C.V: let: } u = \sqrt{x^2+1} \Rightarrow du = \frac{x}{\sqrt{x^2+1}} \cdot dx. \text{ so: } D = \int (u^2-1) \cdot du = \frac{1}{3} u^3 - x + C$$

$$\Rightarrow D = \frac{1}{3} (\sqrt{x^2+1})^3 - x + C = \frac{1}{3} (x^2+1)^{3/2} - x + C.$$

Exercise (5):

$$1) I_1 = \int \cos(\ln(x)) \cdot dx. \text{ let: } u = \ln(x), \text{ so: } I_1 = \int \cos(u) \cdot e^u \cdot du \quad [u = \ln(x) \Rightarrow x = e^u]$$

$$du = \frac{1}{x} dx \Rightarrow dx = x du = e^u \cdot du. \text{ by using I.B.P method we get:}$$

$$\int \cos(u) \cdot e^u \cdot du = \cos(u) \cdot e^u + \int \sin(u) \cdot e^u \cdot du. \text{ by using I.B.P again we get:}$$

$$w = \cos(u) \Rightarrow dw = -\sin(u) \cdot du$$

$$v = e^u \Rightarrow dv = e^u \cdot du.$$

$$\int \sin(u) \cdot e^u \cdot du = \sin(u) \cdot e^u - \int \cos(u) \cdot e^u \cdot du$$

$$w = \sin(u) \Rightarrow dw = \cos(u) \cdot du$$

$$v = e^u \Rightarrow dv = e^u \cdot du$$

$$\text{So: } \int \cos(u) \cdot e^u \cdot du = \cos(u) \cdot e^u + \sin(u) \cdot e^u - \int \cos(u) \cdot e^u \cdot du$$

$$\Rightarrow 2 \int \cos(u) \cdot e^u \cdot du = \cos(u) \cdot e^u + \sin(u) \cdot e^u$$

$$\Rightarrow \int \cos(u) \cdot e^u \cdot du = \frac{1}{2} [e^u (\sin(u) + \cos(u))]$$

$$\text{then we get: } I_1 = \frac{1}{2} [e^{\ln(x)} (\sin(\ln(x)) + \cos(\ln(x))) + C = \frac{1}{2} [x (\sin(\ln(x)) + \cos(\ln(x))) + C$$

$$2) I_2 = \int \arg \cosh(x) = x \cdot \arg \cosh(x) - \int \frac{x \cdot \frac{1}{\sqrt{x^2-1}}}{\sqrt{x^2-1}} dx = x \cdot \arg \cosh(x) - \sqrt{x^2-1} + C$$

$$u = \arg \cosh(x) \Rightarrow du = \frac{1}{\sqrt{x^2-1}} dx$$

$$v = x \Rightarrow dv = 1 dx$$

$$3) I_3 = \int x \cdot \tan^2(x) \cdot dx = x \tan(x) - x^2 - \int \tan(x) - x \cdot dx = x \tan(x) - x^2 + \ln|\cos(x)| + \frac{1}{2} x^2 + C$$

$$u = x \Rightarrow du = 1 \cdot dx$$

$$= x \tan(x) - \frac{1}{2} x^2 + \ln|\cos(x)| + C$$

$$v = \tan(x) - x \Rightarrow dv = \tan^2(x) \cdot dx$$

Exercise (6):

$$1) I_n = \int \frac{dt}{(t^3+1)^n} = \frac{t}{(t^3+1)^n} + 3n \int \frac{t^3+1-1}{(t^3+1)^{n+1}} dt = \frac{t}{(t^3+1)^n} + 3n \int \frac{dt}{(t^3+1)^n} - 3n \int \frac{dt}{(t^3+1)^{n+1}}$$

$$u = \frac{1}{(t^3+1)^n} \Rightarrow du = -3n \cdot t^2 \cdot (t^3+1)^{-(n+1)} dt$$

$$v = t \Rightarrow dv = 1 dt$$

$$\text{we get: } I_{n+1} = \frac{1}{3n} \left[\frac{t}{(t^3+1)^n} + (3n-1) I_n \right]$$

$$2) I = \int \frac{dx}{1+\cos(x)} \text{ we have: } 1+\cos(x) = 2 \cos^2\left(\frac{x}{2}\right), \text{ then: } I = \int \frac{dx}{2 \cos^2\left(\frac{x}{2}\right)} = \int \frac{1/2 \cdot dx}{\cos^2\left(\frac{x}{2}\right)}$$

$$u = \frac{x}{2} \Rightarrow du = \frac{1}{2} \cdot dx$$

$$= \int \frac{du}{\cos^2(u)} = \tan\left(\frac{x}{2}\right) + C$$

Exercise (7):

$$I_1 = \int (\cos x)^{\frac{2024}{2025}} \cdot \sin x \cdot dx \text{ put: } u = \cos(x) \Rightarrow du = -\sin x \cdot dx, \text{ then:}$$

$$\int (\cos x)^{\frac{2024}{2025}} \cdot \sin x \cdot dx = - \int u^{\frac{2024}{2025}} \cdot du = \frac{-u^{\frac{2025}{2025}}}{\frac{2025}{2025}} + C = \frac{-(\cos x)^{\frac{2025}{2025}}}{2025} + C$$

$$I_2 = \int \frac{dx}{x \ln x} \text{ put: } u = \ln(x) \Rightarrow du = \frac{1}{x} dx, \text{ then: } \int \frac{dx}{x \ln(x)} = \int \frac{du}{u} = \ln|u| + C$$

$$= \ln|\ln(x)| + C$$

$$I_3 = \int \frac{dx}{3+e^{-x}} = \int \frac{e^x}{3e^x+1} dx. \text{ put: } t = 3e^x+1 \Rightarrow dt = 3e^x dx \Rightarrow \frac{1}{3} dt = e^x dx, \text{ then:}$$

$$= \int \frac{1}{3} \cdot \frac{dt}{t} = \frac{1}{3} \ln|t| + c = \frac{1}{3} \ln(3e^x+1) + c.$$

$$I_4 = \int \frac{dx}{\sqrt{4x-x^2}} = \int \frac{dx}{\sqrt{4-(x-2)^2}}. \text{ let: } t = x-2 \Rightarrow dt = 1 dx, \text{ then:}$$

$$= \int \frac{dt}{\sqrt{4-t^2}} = \arcsin\left(\frac{t}{2}\right) + c = \arcsin\left(\frac{x-2}{2}\right) + c$$

Exercise (8):

$$a) I_1 = \int \frac{dx}{x^2+2x+5}. \forall x \in \mathbb{R}: I = \int \frac{dx}{x^2+2x+1+4} = \int \frac{dx}{(x+1)^2+4} = \frac{1}{4} \int \frac{dx}{\left(\frac{x+1}{2}\right)^2+1}. \text{ put:}$$

$$u = \frac{x+1}{2} \Rightarrow du = \frac{1}{2} dx \Rightarrow dx = 2 du, \text{ then: } I = \frac{1}{4} \int \frac{2 du}{u^2+1} = \frac{1}{2} \arctan(u) + c$$

$$= \frac{1}{2} \arctan\left(\frac{x+1}{2}\right) + c.$$

$$I_2 = \int \frac{dx}{x^2+3x+1}. \text{ we have: } \Delta = b^2-4ac = 5 > 0. \text{ so: } x_1 = \frac{-3+\sqrt{5}}{2} \text{ and: } x_2 = \frac{-3-\sqrt{5}}{2}$$

$$\text{then: } x^2+3x+1 = \left(x - \frac{3-\sqrt{5}}{2}\right)\left(x - \frac{3+\sqrt{5}}{2}\right) \Rightarrow \frac{1}{x^2+3x+1} = \frac{a}{x - \frac{3-\sqrt{5}}{2}} + \frac{b}{x - \frac{3+\sqrt{5}}{2}}$$

$$\begin{cases} a+b=0 \\ -\frac{3+\sqrt{5}}{2}a - \frac{3-\sqrt{5}}{2}b = 1 \end{cases} \Rightarrow \begin{cases} b = -a \\ -\frac{3-\sqrt{5}}{2}a + \frac{3-\sqrt{5}}{2}a = 1 \end{cases} \Rightarrow a = \frac{-1}{\sqrt{5}} \text{ and: } b = \frac{1}{\sqrt{5}}$$

$$I_2 = \frac{1}{\sqrt{5}} \int \frac{dx}{x - \frac{3-\sqrt{5}}{2}} + \frac{1}{\sqrt{5}} \int \frac{dx}{x - \frac{3+\sqrt{5}}{2}} = \frac{-1}{\sqrt{5}} \ln\left|x - \frac{3-\sqrt{5}}{2}\right| + \frac{1}{\sqrt{5}} \ln\left|x - \frac{3+\sqrt{5}}{2}\right| + c$$

$$= \frac{1}{\sqrt{5}} \left[\ln\left|x - \frac{3+\sqrt{5}}{2}\right| - \ln\left|x - \frac{3-\sqrt{5}}{2}\right| \right] + c = \frac{1}{\sqrt{5}} \ln\left|\frac{2x-3-\sqrt{5}}{2x-3+\sqrt{5}}\right| + c$$

$$I_3 = \int \frac{3x-2}{5x^2-3x+2} dx. \text{ we have: } (5x^2-3x+2)' = 10x-3, \forall x \in \mathbb{R}. \text{ then:}$$

$$3x-2 = \frac{3}{10} \left(10x - \frac{20}{3}\right) = \frac{3}{10} \left(10x-3 + 3 - \frac{20}{3}\right) = \frac{3}{10} \left(10x-3 - \frac{11}{3}\right), \text{ and:}$$

$$5x^2-3x+2 = 5\left[x^2 - \frac{3}{5}x + \frac{2}{5}\right] = 5\left[x^2 - 2 \cdot \frac{3}{5} \cdot \frac{x}{2} + \frac{2}{5}\right] = 5\left[x^2 - 2 \cdot \frac{3}{5} \cdot \frac{x}{2} + \frac{2}{5} + \left(\frac{3}{10}\right)^2 - \left(\frac{3}{10}\right)^2\right]$$

$$= 5\left[\left(x - \frac{3}{10}\right)^2 + \frac{2}{5} - \frac{9}{100}\right] = 5\left[\left(x - \frac{3}{10}\right)^2 + \frac{31}{100}\right]. \text{ then:}$$

$$I_3 = \frac{3}{10} \int \frac{10x-3-\frac{11}{3}}{5x^2-3x+2} dx = \frac{3}{10} \int \frac{10x-3}{5x^2-3x+2} dx - \frac{11}{10} \int \frac{dx}{5x^2-3x+2}. \text{ put:}$$

$$u = x - \frac{3}{10} \Rightarrow du = 1 dx, \text{ thus: } I_3 = \frac{3}{10} \ln|5x^2-3x+2| - \frac{11}{50} \int \frac{dx}{\left(x - \frac{3}{10}\right)^2 + \frac{31}{100}}$$

$$= \frac{3}{10} \ln(15x^2 - 3x + 2) - \frac{11}{50} \int \frac{du}{(u^2 + \frac{31}{100})} = \frac{3}{10} \ln|5x^2 - 3x + 2| - \frac{11}{50} \times \frac{1}{\sqrt{\frac{31}{100}}} \arctg\left(\frac{4}{\sqrt{\frac{31}{100}}}\right) + c$$

$$= \frac{3}{10} \ln|5x^2 - 3x + 2| - \frac{11}{5\sqrt{31}} \arctg\left(\frac{10x-3}{\sqrt{31}}\right) + c.$$

$$I_4 = \int \frac{6x^4 - 5x^3 + 4x^2}{2x^2 - x + 1} dx = \int \frac{(2x^2 - x + 1)(3x^2 - x)}{2x^2 - x + 1} dx = \int (3x^2 - x) dx + \int \frac{x}{2x^2 - x + 1} dx$$

$$\begin{array}{r} 6x^4 - 5x^3 + 4x^2 \\ - 6x^4 - 3x^3 + 3x^2 \\ \hline 0 - 2x^3 + x^2 \\ - 2x^3 + x^2 - x \\ \hline x \end{array} \quad \begin{array}{l} 2x^2 - x + 1 \\ 3x^2 - x \end{array}$$

$$= \int \frac{x}{2x^2 - x + 1} dx + 3 \int x^2 dx - \int x dx$$

$$= \int \left(\frac{4x-1}{4(2x^2 - x + 1)} + \frac{1}{4(2x^2 - x + 1)} \right) dx + 3 \int x^2 dx - \int x dx$$

$$= \frac{1}{4} \int \frac{4x-1}{2x^2 - x + 1} dx + \frac{1}{4} \int \frac{dx}{2x^2 - x + 1} + 3 \int x^2 dx - \int x dx$$

put: $u = 2x^2 - x + 1 \Rightarrow du = 4x - 1 dx$, then:

$$= \frac{1}{4} \ln|2x^2 - x + 1| + \frac{1}{4} \int \frac{dx}{2x^2 - x + 1} + 3 \int x^2 dx - \int x dx = \frac{1}{4} \ln|2x^2 - x + 1| + \frac{1}{4} \int \frac{dx}{(2x - \frac{1}{2})^2 + \frac{7}{8}} + 3 \int x^2 dx - \int x dx.$$

we have: $\frac{1}{4} \int \frac{dx}{(2x - \frac{1}{2})^2 + \frac{7}{8}} = \frac{1}{4} \int \frac{dx}{\frac{7}{8} \left[\frac{(2x - \frac{1}{2})^2}{\frac{7}{8}} + 1 \right]} = \frac{2}{7} \int \frac{dx}{\left(\frac{2x - \frac{1}{2}}{\sqrt{7/8}} \right)^2 + 1}$ put: $u = \frac{2x - \frac{1}{2}}{\sqrt{7/8}}$

then: $I_4 = \frac{1}{4} \ln|2x^2 - x + 1| + \frac{2}{7} \cdot \arctg\left(\frac{2x - \frac{1}{2}}{\sqrt{7/8}}\right) + 3 \int x^2 dx - \int x dx$

$$= \frac{1}{4} \ln|2x^2 - x + 1| + \frac{2}{7} \cdot \arctg\left(\frac{2x - \frac{1}{2}}{\sqrt{7/8}}\right) + x^3 - \frac{1}{2} x^2 + c.$$

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b) $I_1 = \int \frac{x+2}{x^2 - 3x + 4} dx$ we have: $x+2 = \frac{1}{2}(2x-3) + \frac{7}{2}$, then: $I_1 = \int \frac{\frac{1}{2}(2x-3) + \frac{7}{2}}{x^2 - 3x + 4} dx$

$$= \frac{1}{2} \int \frac{2x-3}{x^2 - 3x + 4} + \frac{7}{2} \int \frac{dx}{x^2 - 3x + 4} = \frac{1}{2} \ln|x^2 - 3x + 4| + \frac{7}{2} \int \frac{dx}{(x - \frac{3}{2})^2 + \frac{7}{4}}$$

$$\frac{7}{2} \int \frac{dx}{(x - \frac{3}{2})^2 + \frac{7}{4}} = \frac{7}{2} \int \frac{dx}{\frac{7}{4} \left[\frac{(x - \frac{3}{2})^2}{\frac{7}{4}} + 1 \right]} = 2 \int \frac{dx}{\left(\frac{x - \frac{3}{2}}{\sqrt{7/4}} \right)^2 + 1}$$
 put: $u = \frac{x - \frac{3}{2}}{\sqrt{7/4}}$, then:

$$I_1 = \frac{1}{2} \ln|x^2 - 3x + 4| + 2 \arctg\left(\frac{x - \frac{3}{2}}{\sqrt{7/4}}\right) + c$$

$I_2 = \int \sin^3(x) \cdot \cos^3(x) dx = \int \sin^2(x) \cdot \cos^2(x) \cdot \cos(x) dx$ put: $t = \sin(x) \Rightarrow dt = \cos(x) dx$

and we have: $\cos^2(x) = 1 - \sin^2(x) = 1 - t^2$, then: $I_2 = \int \sin^2(x) \cdot (1 - \sin^2(x)) \cdot dt$

$$= \int t^2 (1 - t^2) dt = \int t^2 - t^{10} dt = \frac{1}{3} t^3 - \frac{1}{11} t^{11} + c = \frac{1}{3} \sin^3(x) - \frac{1}{11} \sin^{11}(x) + c.$$

$I_3 = \int \frac{1}{\sin(x)} dx$ put: $t = \tg\left(\frac{x}{2}\right) \Rightarrow x = 2 \arctg(t) \Rightarrow dx = \frac{2}{1+t^2} dt$, and we

have: $\sin(x) = \frac{2t}{1+t^2}$, $\cos(x) = \frac{1-t^2}{1+t^2}$, then:

$$I_3 = \int \frac{1}{\cancel{2t} \cdot \cancel{1+t^2}} dt = \int \frac{1}{t} dt = \ln|t| + C = \ln\left|\tan\left(\frac{x}{2}\right)\right| + C$$

$$I_4 = \int \frac{3 - \sin(x)}{2\cos(x) + 3\tan(x)} dx = - \int \frac{\sin(x) - 3}{\frac{2\cos^2(x) + 3\sin(x)}{\cos(x)}} = - \int \cos(x) \frac{\sin(x) - 3}{2 - 2\sin^2(x) + 3\sin(x)} dx$$

put: $u = \sin(x) \Rightarrow du = \cos(x) \cdot dx$, then: $I_4 = - \int \frac{u-3}{2-2u^2+3u} du = - \int \frac{u-3}{2u^2-3u-2} du$

we have: $\frac{u-3}{2u^2-3u-2} = \frac{u-3}{(u-2)(2u+1)} = \frac{a}{u-2} + \frac{b}{2u+1} = \frac{2au+a+bu-2b}{2u^2-3u-2}$

$$\begin{cases} 2a+b=1 & \text{---(1)} \\ a-2b=-3 & \text{---(2)} \end{cases} \Rightarrow \begin{cases} \text{from (1) we get: } b=1-2a \\ \text{and from (2): } a-2(1-2a)=-3 \Rightarrow 5a=-1 \end{cases} \Rightarrow \begin{cases} \text{then: } a = -\frac{1}{5} \\ \text{and: } b = \frac{7}{5} \end{cases}$$

thus: $I_4 = -\frac{1}{5} \int \frac{1}{u-2} du + \frac{7}{5} \int \frac{1}{2u+1} du = -\frac{1}{5} \ln|u-2| + \frac{7}{10} \ln|2u+1| + C$

$$= -\frac{1}{5} \ln|\sin(x)-2| + \frac{7}{10} \ln|2\sin(x)+1| + C.$$

Exercise (9):

1) $I_1 = \int (x + \sqrt{x}) dx = \int x dx + \int \sqrt{x} dx = \int x dx + \int (x)^{1/2} dx = \frac{1}{2} x^2 + \frac{2}{3} x^{3/2} + C$

2) $I_2 = \int \frac{dx}{\sqrt[4]{x}} = \int \frac{dx}{(x)^{1/4}} = \int (x)^{-1/4} dx = \frac{4}{3} x^{3/4} + C.$

3) $I_3 = \int \left(\frac{3}{\sqrt{x}} - \frac{x\sqrt{x}}{4} \right) dx = 3 \int \frac{1}{\sqrt{x}} dx - \frac{1}{4} \int x\sqrt{x} dx = 3 \int x^{-1/2} dx - \frac{1}{4} \int x^{3/2} dx = 6x^{1/2} - \frac{1}{10} x^{5/2} + C$

Exercise (10):

a) $I_1 = \int \frac{dx}{(\sin 3x)^2}$. put: $u = 3x \Rightarrow du = 3 dx \Rightarrow dx = \frac{1}{3} du$, then:

$$\begin{aligned} &= \int \frac{du}{3 \sin^2(u)} = \frac{1}{3} \int \frac{1}{\sin^2(u)} du = \frac{1}{3} \int \frac{\sin^2(u) + \cos^2(u)}{\sin^2(u)} du = \frac{1}{3} \int 1 + \cot^2(u) du \\ &= \frac{1}{3} \cot(u) + C = \frac{1}{3} \cot(3x) + C \end{aligned}$$

$I_2 = \int \frac{dx}{5-2x}$. put: $u = 5-2x \Rightarrow du = -2 dx$, then: $I_2 = \frac{-1}{2} \int \frac{-2}{5-2x} dx = \frac{-1}{2} \ln|5-2x| + C$

$I_3 = \int (\sin x)^2 \cdot \cos(x) dx$. put: $u = \sin(x) \Rightarrow du = \cos(x) dx$, then: $I_3 = \int u^2 \cdot du$

$$= \frac{1}{3} u^3 + c = \frac{1}{3} (\sin(x))^3 + c$$

b) $I_1 = \int \frac{\sin(2x)}{(1+\cos(2x))^2} dx$. put: $u = 1 + \cos(2x) \Rightarrow du = -2\sin(2x) dx$, then:

$$= \frac{-1}{2} \int \frac{1}{u^2} du = \frac{-1}{2} \times -\frac{1}{u} + c = \frac{1}{2u} + c = \frac{1}{2(1+\cos(2x))} + c$$

$I_2 = \int \frac{\arctan(x)}{1+x^2} dx$. put: $u = \arctan(x) \Rightarrow du = \frac{1}{x^2+1} dx$, then:

$$= \int u du = \frac{1}{2} u^2 + c = \frac{1}{2} \arctan(x) + c$$

$I_3 = \int \frac{dx}{\sqrt{1-x^2} \arcsin(x)}$. put: $u = \arcsin(x) \Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$, then:

$$= \int \frac{1}{u} du = \ln|u| + c = \ln|\arcsin(x)| + c$$

c) $I_1 = \int \frac{\cos(\ln(x))}{x} dx$. put: $u = \ln(x) \Rightarrow du = \frac{1}{x} dx$, then:

$$= \int \cos(u) du = \sin(u) + c = \sin(\ln(x)) + c$$

$I_2 = \int \frac{e^x}{\sqrt{1-e^{2x}}} dx$. put: $u = e^x \Rightarrow du = e^x dx$, then: $I_2 = \int \frac{1}{\sqrt{1-u^2}} du$.

$$= \arcsin(u) + c = \arcsin(e^x) + c$$

$I_3 = \int \frac{dx}{1+2x^2}$. put: $u = \sqrt{2}x \Rightarrow du = \sqrt{2} dx$, then: $I_3 = \frac{1}{\sqrt{2}} \int \frac{1}{u^2+1} du = \frac{1}{\sqrt{2}} \arctan(u) + c$

$$= \frac{1}{\sqrt{2}} \arctan(\sqrt{2}x) + c$$

d) $I_1 = \int \frac{dx}{4-9x^2} = \int \frac{1}{(2)^2 - (3x)^2} dx$. put: $u = 3x \Rightarrow du = 3 dx$, then:

$$= \frac{1}{3} \int \frac{1}{(2)^2 - u^2} dt = \frac{1}{6} \cdot \frac{1}{2} \ln \left| \frac{2+u}{2-u} \right| + c = \frac{1}{12} \ln \left| \frac{2+3x}{2-3x} \right| + c .$$

$I_2 = \int \frac{dx}{\sqrt{1-3x^2}}$. put: $u = \sqrt{3}x \Rightarrow du = \sqrt{3} dx$, then: $I_2 = \frac{1}{\sqrt{3}} \int \frac{1}{\sqrt{1-u^2}} dx = \frac{1}{\sqrt{3}} \arcsin(u) + c$

$$= \frac{1}{\sqrt{3}} \arcsin(\sqrt{3}x) + c$$

$I_3 = \int \frac{dx}{2(\sin x)^2 + 3(\cos x)^2} = \int \frac{1}{2(1-\cos^2(x)) + 3\cos^2(x)} dx = \int \frac{dx}{2-2\cos^2(x) + 3\cos^2(x)}$

$$= \int \frac{1}{2+\cos^2(x)} dx$$

1st method: put: $u = \tan(x) \Rightarrow du = \frac{1}{\cos^2(x)} dx = 1 + \tan^2(x) dx \Rightarrow dx = \frac{du}{1+\tan^2(x)}$

$\Rightarrow dx = \frac{du}{1+u^2}$ and we have: $\cos^2(x) = \frac{1}{1+\tan^2(x)}$, then:

$I_3 = \int \frac{1}{2+1/(1+u^2)} \left(\frac{du}{1+u^2} \right) = \int \frac{1}{2u^2+3} du = \frac{1}{\sqrt{2}} \int \frac{\sqrt{2}}{(\sqrt{2}u)^2 + (\sqrt{3})^2} = \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{3}} \arctan\left(\frac{\sqrt{2}u}{\sqrt{3}}\right) + c .$

$$= \frac{1}{\sqrt{6}} \arctan\left(\frac{\sqrt{2}u}{\sqrt{3}}\right) + c = \frac{1}{\sqrt{6}} \arctan\left(\frac{\sqrt{2}\tan x}{\sqrt{3}}\right) + c.$$

2nd method: we know that: $\cos^2(x) = \frac{\cos(2x)+1}{2}$, then:

$$I_3 = \int \frac{1}{2 + \frac{\cos(2x)+1}{2}} dx = \int \frac{1}{\frac{4+\cos(2x)+1}{2}} dx = \int \frac{2}{5+\cos(2x)} dx. \text{ put: } u=2x \Rightarrow du=2dx,$$

$$\text{then: } I_3 = \int \frac{1}{5+\cos(u)} du \text{ and we have: } \cos(u) = \frac{1-t^2}{1+t^2} \text{ and } t = \tan\left(\frac{u}{2}\right) \Rightarrow du = \frac{2dt}{1+t^2}$$

$$\text{so: } \int \frac{\frac{2}{1+t^2} dt}{5 + \frac{1-t^2}{1+t^2}} = \int \frac{\frac{2}{1+t^2}}{\frac{5+5t^2+1-t^2}{1+t^2}} dt = \int \frac{2}{6+4t^2} dt = \frac{2}{6} \int \frac{1}{\left(1+\left(\frac{2t}{\sqrt{6}}\right)^2\right)} dt$$

$$= \frac{2}{6} \cdot \frac{\sqrt{6}}{2} \int \frac{dv}{1+v^2} = \frac{\sqrt{6}}{6} \cdot \arctan(v) + c = \frac{\sqrt{6}}{6} \arctan\left(\frac{2\tan(x)}{\sqrt{6}}\right) + c.$$

Exercise (11):

$$I_1 = \int \arcsin(x) \cdot dx = x \cdot \arcsin(x) - \int \frac{x}{\sqrt{1-x^2}} dx. \text{ put: } u=1-x^2 \Rightarrow du=-2x dx, \text{ then:}$$

$$u = \arcsin(x) \Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$$

$$v = x \Rightarrow dv = 1 dx$$

$$= x \cdot \arcsin(x) + \frac{1}{2} \int \frac{1}{\sqrt{u}} du = x \cdot \arcsin(x) + \frac{1}{2} \int (u)^{-1/2} du$$

$$= x \cdot \arcsin(x) - \frac{1}{2} \cdot 2 (u)^{1/2} + c = x \cdot \arcsin(x) - \sqrt{u} + c.$$

$$= x \cdot \arcsin(x) - \sqrt{1-x^2} + c.$$

$$I_2 = \int x \cdot \arctan(x) \cdot dx = \frac{x^2}{2} \cdot \arctan(x) - \int \frac{x^2}{2} \cdot \frac{1}{1+x^2} dx = \frac{x^2}{2} \arctan(x) - \frac{1}{2} \int \frac{x^2-1+1}{1+x^2} dx$$

$$u = \arctan(x) \Rightarrow du = \frac{1}{1+x^2} dx$$

$$v = \frac{1}{2} x^2 \Rightarrow dv = x dx$$

$$= \frac{x^2}{2} \arctan(x) - \frac{1}{2} \int \frac{x^2-1}{x^2+1} - \frac{1}{1+x^2} dx$$

$$= \frac{x^2}{2} \arctan(x) - \frac{1}{2} \int 1 dx + \frac{1}{2} \int \frac{1}{1+x^2} dx$$

$$= \frac{x^2}{2} \arctan(x) - \frac{1}{2} x + \frac{1}{2} \arctan(x) + c = \frac{x^2 \arctan(x) - x + \arctan(x)}{2} + c.$$

$$I_3 = \int \frac{x \arcsin x}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2} \cdot \arcsin(x) + \int \frac{\sqrt{1-x^2}}{\sqrt{1-x^2}} dx = -\sqrt{1-x^2} \cdot \arcsin(x) + \int 1 dx$$

$$u = \arcsin(x) \Rightarrow du = \frac{1}{\sqrt{1-x^2}} dx$$

$$v = -\sqrt{1-x^2} \Rightarrow dv = \frac{x}{\sqrt{1-x^2}} dx$$

$$= x \cdot \sqrt{1-x^2} \cdot \arcsin(x) + c.$$

Exercise (12): primitives of the form $\int \frac{Ax+B}{\sqrt{ax^2+bx+c}} dx$

$$a \left[\left(x + \frac{b}{2a} \right)^2 + \frac{4ac-b^2}{4a^2} \right]$$

you can use directly the rule

$$I_1 = \int \frac{dx}{\sqrt{2-3x-4x^2}} = \int \frac{dx}{\sqrt{-4x^2-3x+2}}. \text{ we have:}$$

canonical form

$$-4x^2 - 3x + 2 = -4 \left(x^2 + \frac{3}{4}x - \frac{1}{2} \right) = -4 \left(x^2 + 2 \cdot \frac{3}{8}x + \frac{9}{64} - \frac{9}{64} - \frac{1}{2} \right) = -4 \left[\left(x + \frac{3}{8} \right)^2 - \frac{41}{64} \right], \text{ then:}$$

$$I_1 = \int \frac{dx}{\sqrt{-4 \left[\left(x + \frac{3}{8} \right)^2 - \frac{41}{64} \right]}} = \frac{1}{2} \int \frac{dx}{\sqrt{\left(\frac{\sqrt{41}}{8} \right)^2 - \left(x + \frac{3}{8} \right)^2}}. \text{ let: } t = x + \frac{3}{8} \Rightarrow dt = 1 dx, \text{ then:}$$

$$I_1 = \frac{1}{2} \int \frac{dt}{\sqrt{\left(\frac{\sqrt{41}}{8} \right)^2 - t^2}} = \frac{1}{2} \arcsin \left(\frac{t}{\frac{\sqrt{41}}{8}} \right) + C = \frac{1}{2} \arcsin \left(\frac{8x+3}{\sqrt{41}} \right) + C$$

$$I_2 = \int \frac{1}{\sqrt{1+x+x^2}} dx = \int \frac{dx}{\sqrt{x^2+x+1}} = \int \frac{dx}{\sqrt{\left(x + \frac{1}{2} \right)^2 + \frac{3}{4}}}. \text{ let: } t = x + \frac{1}{2} \Rightarrow dt = 1 dx, \text{ then:}$$

$$I_2 = \int \frac{dt}{\sqrt{t^2 + \left(\frac{\sqrt{3}}{2} \right)^2}} = \operatorname{arcsinh} \left(\frac{t}{\frac{\sqrt{3}}{2}} \right) + C = \operatorname{arcsinh} \left(\frac{2x+1}{\sqrt{3}} \right) + C = \operatorname{arcsinh} \left(\frac{2x+1}{\sqrt{3}} \right) + C.$$

$$I_3 = \int \frac{x+3}{\sqrt{4x^2+4x+3}} dx. \text{ we have: } (4x^2+4x+3)' = 8x+4, \text{ then: } x+3 = \frac{1}{8}(8x+24) = \frac{1}{8}(8x+20+4),$$

$$\text{so: } I_3 = \int \frac{1}{8} \left(\frac{8x+4}{\sqrt{4x^2+4x+3}} + \frac{20}{\sqrt{4x^2+4x+3}} \right) dx = \frac{1}{4} \sqrt{4x^2+4x+3} + \frac{5}{2} \int \frac{dx}{\sqrt{4 \left(x^2+x+\frac{3}{4} \right)}}$$

$$= \frac{1}{4} \sqrt{4x^2+4x+3} + \frac{5}{4} \int \frac{dx}{\sqrt{\left(x + \frac{1}{2} \right)^2 + \frac{1}{2}}} = \frac{1}{4} \sqrt{4x^2+4x+3} + \frac{5}{4} \int \frac{dx}{\sqrt{\frac{1}{2} \left(\frac{2x+1}{\frac{1}{\sqrt{2}} \right)^2 + 1}}$$

$$= \frac{1}{4} \sqrt{4x^2+4x+3} + \frac{5}{4} \cdot \sqrt{2} \int \frac{dx}{\sqrt{\left(\frac{2x+1}{\sqrt{2}} \right)^2 + 1}}. \text{ let: } t = \frac{2x+1}{\sqrt{2}} \Rightarrow dx = \frac{\sqrt{2}}{2} dt, \text{ then:}$$

$$I_3 = \frac{1}{4} \sqrt{4x^2+4x+3} + \frac{5\sqrt{2}}{4} \cdot \frac{\sqrt{2}}{2} \int \frac{dt}{\sqrt{t^2+1}} = \frac{1}{4} \sqrt{4x^2+4x+3} + \frac{5}{4} \operatorname{arcsinh} \left(\frac{2x+1}{\sqrt{2}} \right) + C.$$

$$I_4 = \int \frac{x+3}{\sqrt{-4x^2+4x+3}} dx. \text{ we have: } (-4x^2+4x+3)' = -8x+4, \text{ then: } x+3 = \frac{-1}{8}(-8x-24) = \frac{-1}{8}(-8x+4-28)$$

$$\text{so: } I_4 = \int \frac{-1}{8} \left(\frac{-8x+4}{\sqrt{-4x^2+4x+3}} - \frac{28}{\sqrt{-4x^2+4x+3}} \right) dx = \frac{-1}{8} \int \left(\frac{-8x+4}{\sqrt{-4x^2+4x+3}} - \frac{28}{\sqrt{-4x^2+4x+3}} \right) dx$$

$$= \frac{-1}{8} \int \frac{-8x+4}{\sqrt{-4x^2+4x+3}} + \frac{7}{2} \int \frac{dx}{\sqrt{-4 \left[\left(x - \frac{1}{2} \right)^2 - 1 \right]}} = \frac{-1}{4} \sqrt{-4x^2+4x+3} + \frac{7}{4} \int \frac{dx}{\sqrt{1 - \left(x - \frac{1}{2} \right)^2}}.$$

$$\text{let: } t = x - \frac{1}{2} \Rightarrow dt = 1 dx, \text{ then: } I_4 = \frac{-1}{4} \sqrt{-4x^2+4x+3} + \frac{7}{4} \int \frac{1}{\sqrt{1-t^2}} dx$$

$$= \frac{-1}{4} \sqrt{-4x^2+4x+3} + \frac{7}{4} \arcsin(t) + C = \frac{-1}{4} \sqrt{-4x^2+4x+3} + \frac{7}{4} \arcsin \left(\frac{2x-1}{2} \right) + C$$

Exercise (14): $\int R(x^{\frac{1}{n}}, x^{\frac{1}{l}}, x^{\frac{1}{k}}, \dots), \alpha = \operatorname{LCM}(n, l, k, \dots)$

$$\text{a) } I_1 = \int \frac{\sqrt{x^3} - \sqrt[3]{x}}{6\sqrt[4]{x}} dx = \frac{1}{6} \int \frac{x^{\frac{3}{2}} - x^{\frac{1}{3}}}{x^{\frac{1}{4}}} dx. \text{ put: } \alpha = \operatorname{LCM}(2, 3, 4) = 12. \text{ let: } x = t^{12} \Rightarrow dx = 12t^{11} dt,$$

$$\text{then: } I_1 = \frac{1}{6} \int \frac{(t^{12})^{3/2} - (t^{12})^{1/3}}{(t^{12})^{1/4}} \cdot 12t^{11} dt = \frac{1}{6} \int \frac{t^{18} - t^4}{t^3} \cdot 12t^{11} dt = 2 \int \frac{t^3(t^{15} - t)}{t^3} t^{11} dt$$

$$= 2 \int (t^{15} - t) t^{11} dt = 2 \int (t^{26} - t^{12}) dt = 2 \left(\frac{t^{27}}{27} - \frac{t^{13}}{13} \right) + C = 2 \left(\frac{x^{9/4}}{27} - \frac{x^{13/12}}{13} \right) + C.$$

$$I_2 = \int \sqrt{\frac{1-x}{1+x}} \cdot \frac{1}{x} dx = \int \left(\frac{1-x}{1+x} \right)^{1/2} \cdot \frac{1}{x} dx. \text{ let: } \sqrt{\frac{1-x}{1+x}} = t \Rightarrow \frac{1-x}{1+x} = t^2 \text{ (to avoid } |t| \text{), then:}$$

$$1-x = t^2 + t^2 x \Rightarrow 1-t^2 = x(t^2+1) \Rightarrow x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{-2t(1+t^2) - 2t(1-t^2)}{(1+t^2)^2} dt = \frac{-2t - 2t^3 - 2t + 2t^3}{(1+t^2)^2} dt$$

$$\Rightarrow dx = \frac{-4t}{(1+t^2)^2} dt, \text{ so: } x = \frac{1-t^2}{1+t^2} \Rightarrow dx = \frac{-4t}{(1+t^2)^2} dt, \text{ then:}$$

$$I_2 = -4 \int t \cdot \frac{1+t^2}{1-t^2} \cdot \frac{t}{(1+t^2)^2} dt = \int \frac{t^2}{(1+t^2)(1-t^2)} dt = \int \left(\frac{at+b}{1+t^2} + \frac{c}{1-t} + \frac{d}{1+t} \right) dt$$